

Status-Relational Entropy (SRE) Dynamics: Mapping Prerequisites

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Version: 1.6 (Revised for Multi-Scale Homomorphic Mapping)

Version Note: This revision reconstructs the ontological constraints of mapping and no longer presupposes a direct one-to-one projection from the underlying layer to the rendering layer. Drawing on the ontological features of the hierarchical dissipative self-organizing binary-network substrate, it introduces **multi-scale rigid boundaries-homomorphic mapping** as a necessary precondition for valid mapping. Terminology is fully consistent with the SRE Axioms v1.6, Reciprocal-Measurement paper, and Causal-Depth-and-Spatial-Indexing manuscript suite. This document does not address the ultimate origin of causal differences; it characterizes only the intrinsic constraints and feasible pathways for instantiated projection from causal space to the physical rendering layer.

I. Intrinsic Ontological Constraints of Mapping

SRE-Dynamics takes **a set of quantum-evidential causal nodes with asynchronous informational differences** as the underlying input for mapping. Asynchronous evolution already proceeds inside the underlying causal space and numerous causal interactions exist, yet the vast majority of these interactions remain in the **pending-for-instantiation state** and have not yet generated macroscopically observable spatiotemporal states-of-affairs.

The underlying causal network is subject to two ontological constraints that render a pure direct isomorphic mapping fundamentally unfeasible:

- 1. Decoherence-Dissipation Constraint:** Asynchronous activation and local frustration effects exist within the underlying network. A large number of degrees-of-freedom undergo decoherence and dormancy. Evolution is accompanied by irreversible informational dissipation, making lossless upward transmission of every logical degree-of-freedom from the substrate impossible.
- 2. Emergent Ontological Ultraviolet-Boundary Constraint:** Limited by instantiation-cost thresholds, numerous minute underlying logical differences cannot pass instantiation screening and cannot be converted into spatiotemporal states-of-affairs within the physical rendering layer.

An isomorphic mapping requires bijective one-to-one correspondence and complete preservation of all underlying details. Owing to the two constraints above, direct bijective projection from underlying causal space onto the macroscopic physical rendering layer cannot be accomplished. Mapping cannot be performed unconditionally and must be subject to scale-dependent preconditions.

II. Multi-Scale Rigid Boundaries: Pre-Requisite Thresholds for Valid Mapping

To accomplish information-physical instantiated mapping, **rigid coherence truncation boundaries** must be established at the micro-, meso-, and macroscopic scales respectively.

These rigid boundaries are not material physical boundaries. Instead they are statistical-topological screening thresholds: they filter out degrees-of-freedom that are unstable, strongly fluctuating, and unable to satisfy instantiation conditions at the given scale; only causal structures and relational associations possessing statistical stability at that scale are retained.

- **Micro-scale rigid boundary:** Select the high-topological-coherence core region of the system, screen out strongly fluctuating mesoscopic bifurcation degrees-of-freedom outside the core, and preserve locally stable topological-manifold behaviour.
- **Meso-scale rigid boundary:** Set cascade-amplification thresholds to filter extreme nonlinear stochastic bifurcations. Retain only statistically self-consistent causal associations in the ensemble-averaged sense and discard isolated accidental stochastic fluctuations.
- **Macro-scale rigid boundary:** Merge degrees-of-freedom that have undergone long-range decoherence. Extract only collectively emergent observable quantities without tracking detailed states of individual underlying nodes.

In essence, rigid boundaries constitute scale-specific instantiation-selection criteria. Each hierarchical level defines which causal structures may participate in upward mapping and which degrees-of-freedom shall be compressed and screened out.

III. Homomorphic Mapping under Multi-Scale Constraints

When the three sets of rigid boundaries are simultaneously enforced, a **full homomorphic mapping** can be established from causal space to the physical rendering layer.

A homomorphic mapping does not demand one-to-one correspondence between substrate-layer and rendering-layer elements. It permits many-to-one coarse-graining and informational compression, while preserving the core dynamical structures of the system: dissipation-compensation duality, causal-temporal chains, coherence-degradation laws of topological residual manifolds, and BBP spectral phase-transition behaviour.

Based on this mapping framework, cross-scale physical deductions can be carried out, covering: gluon-sea explanations for hadron mass and reconstruction of Maxwell's equations at the quantum scale; multi-scale modelling of the Navier-Stokes equations and VASP-style first-principles analyses at the molecular scale; Earth-interior stress calculations and stellar-spin dynamics at the macroscopic scale. Full derivations and numerical examples are provided in other works within this series.

Within this mapping mechanism:

1. Numerous dormant, decoherent microscopic degrees-of-freedom from the substrate are merged and compressed via homomorphic mapping and are not output as independent physical states-of-affairs at higher layers.
2. Stable causal interactions that pass screening undergo dissipation-compensation accounting via Möbius topological-residual-manifold-mediated reciprocal measurement. After satisfying instantiation conditions, they transition from the pending-for-instantiation state into the physical rendering layer and produce observable phenomena of spacetime, matter, and interactions.
3. Neither time nor space are pre-existing background stages at the fundamental level. Both are derivative products emergent from dissipation-compensation book-keeping after homomorphic-mapping output.

Multidimensional Scaling (MDS) dimensionality reduction in numerical simulations provides a concrete computational realisation of this homomorphic-mapping concept: it discards abundant stochastic details of the underlying binary network and homomorphically projects high-dimensional topological relations onto lower-dimensional emergent geometric manifolds.

IV. Logical Primacy of Reciprocal Measurement

Within the mapping workflow, **reciprocal measurement mediated by light-residual topological manifolds holds logical primacy**.

Time and space coordinates do not exist prior to measurement interactions. On the contrary: one complete reciprocal-measurement event represents one full round of dissipation-evaluation and compensation-settlement. Each settlement contributes one new increment to temporal book-keeping and simultaneously updates topological-book-keeping results for spatial metrics.

Durational perception and spatial distance inside the rendering layer are macroscopic manifestations after homomorphic-mapping of large-numbers of reciprocal-measurement events. Conformal-gauge covariance further guarantees the invariance of locally measured speed-of-light inside the rendering layer.

V. Conclusion

The SRE Mapping Prerequisites define the domain-of-applicability for the framework. This work does not answer the question of the ultimate origin of causal differences. Instead it articulates the constraints required for mapping to occur.

Direct isomorphic projection from the underlying substrate to the physical rendering layer does not exist. Only under the constraints of micro-meso-macro three-level rigid coherence-truncation boundaries can homomorphic instantiated mapping from causal space to the physical rendering layer be realised. Spacetime, matter, and interactions are all emergent outputs of this homomorphic-mapping procedure.

Core-Concept Memorandum (Version-traceability, internal-document use only)

1. Discarded assumptions inherited from older versions: absolute-symmetry zero-void initial state, single initial pulse, distance equivalent to logical-hop count, Planck-scale quantities treated as hardware granularity of underlying network, unconditional direct mapping.
 2. New core logical components: decoherence-dissipation plus emergent ontological ultraviolet boundaries of the substrate render isomorphic mapping infeasible; multi-scale rigid coherence-truncation boundaries as pre-conditions; constrained homomorphic mapping preserving core dynamics; logical primacy of reciprocal measurement.
 3. Unified terminology inventory: emergent ontological ultraviolet boundary, pending-for-instantiation state, physical rendering layer, dissipation-compensation duality, Möbius topological-residual manifold, reciprocal measurement, BBP spectral phase transition, rigid coherence-truncation boundary, homomorphic mapping.
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